

Loop Quantum Gravity and String Theory

John Joseph Marchetta

J. J Marchetta, A. Kar, and G. Cleaver

Dept. of Physics &

Early Universe Cosmology & Strings Division

Center for Astrophysics, Space Physics, & Engineering Research

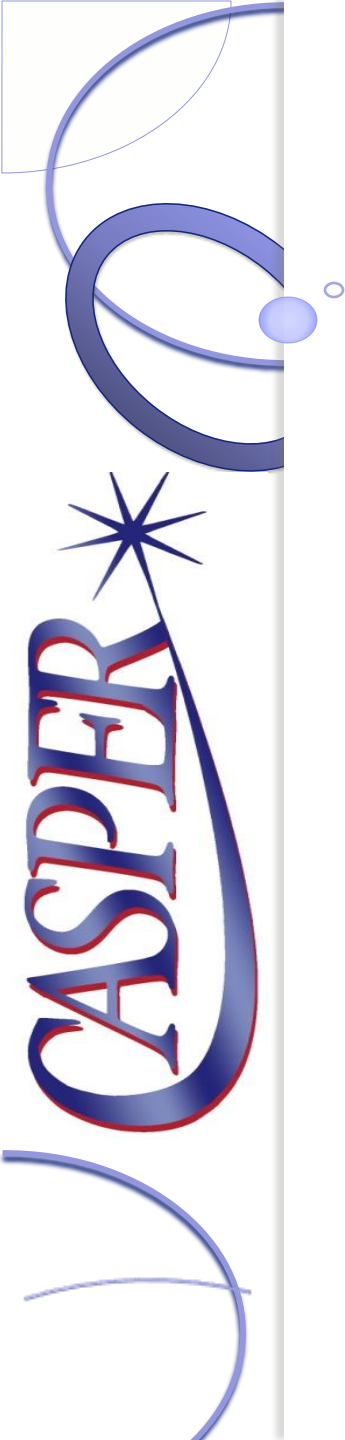
Baylor University, Waco, Texas

April APS Twin Cities, 2023

April 17, 2023



Baylor University





Outline

- Intro to LQG/ loop quantizing Nambu-Goto string (Thiemann '06)
- Top. string theory/ quantum geometry/ Top M theory (Vafa et'al '03-'04)
- Correspondence between Spacelike M-branes and LQG states (Addazi, Marciano '18)
- Future outlooks

Quantizing Loop Algebra (Review by Dona '10)

- ADM gravity config. space $(q_{ab}, \pi^{ab}) \rightarrow (A_a^i, E_i^a)$
- 3 types of constraints, 2 Hamiltonian, 1 Gauss

$$H_a = \frac{1}{\gamma} F_{ab}^j E_j^b - \frac{1 + \gamma^2}{\gamma} K_a^i G_i,$$

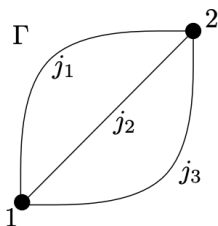
$$G_i = D_a E_i^a$$

$$H = \left[F_{ab}^j - (\gamma^2 + 1) \varepsilon_{jmn} K_a^m K_b^n \right] \frac{\varepsilon_{jkl} E_k^a E_l^b}{\det E} + \frac{1 + \gamma^2}{\gamma} G^i \partial_a \frac{E_i^a}{\det E}.$$

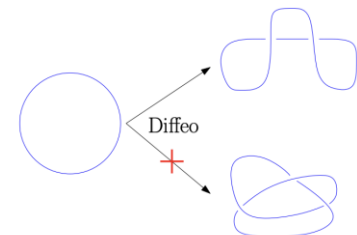
- *-algebra holonomy flux (Unique diff invariant D=3 rep)

$$h_e(A) = \mathcal{P} \exp \left(\int_0^t dt' A_e(t') \right) \quad E_i(S) = \int_S d^2 s \, n_a(s) E_i^a(x(s)) \quad \{E_f(S), h_e(A)\} = \frac{\sigma(S, e)}{2} h(e_1) \tau_i h(e_2) f^i(S \cap e)$$

- Space of spin networks, not Fock (label by knot class)

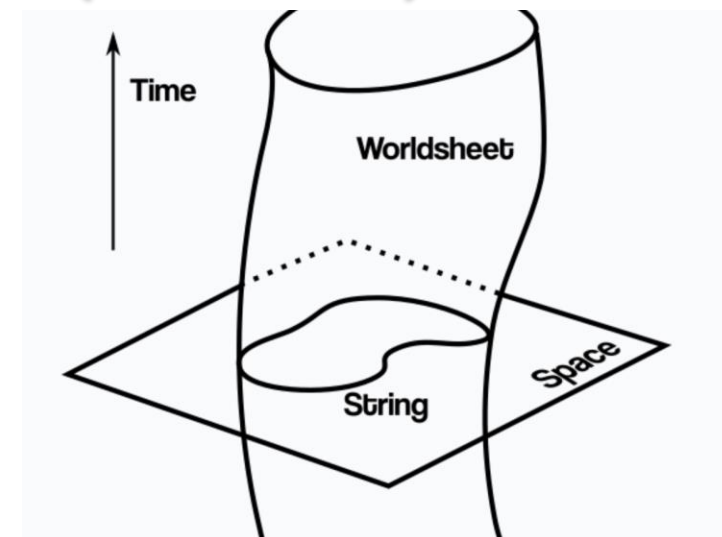


$$\psi(h_L) = \sum_{j_1 \dots j_L} C_{j_1 \dots j_L} i_1^{m_1 m_2 m_3} \dots i_N^{m_{L-2} m_{L-1} m_L} D_{m_1 n_1}^{j_1}(h_{e_1}) \dots D_{m_L n_L}^{j_L}(h_{e_L})$$



Loop quantizing Nambu-Goto (TT '06)

- String is a 2D field theory with diff. gauge symmetry.
- Hypersurface Deformation alg has no Phase space dependent structure const.
- Analogs of holonomy flux alg. and spin networks



$$Y_{\pm}^{\mu} := \eta^{\mu\nu} \pi_{\nu} \pm X^{\mu'}$$

$$Y_{\pm}(f) := \int_{S^1} dx f_{\mu} Y_{\pm}^{\mu}$$

$$\hat{W}_{\epsilon}^k(I) := e^{i\hat{Y}_{\epsilon}^k(I)}$$

- Deviation between loop and Fock constructions:

$$\omega_F(W(f)) = e^{-\frac{1}{4}\langle f|f \rangle}$$

$$\omega_P(W(f)) = \begin{cases} 1 & \text{if } f = 0 \\ 0 & \text{else.} \end{cases}$$

Fock v.s. Loop (HP '06)

- GNS states: $\omega_F(W(f)) = e^{-\frac{1}{4}\langle f|f \rangle}$ $\omega_P(W(f)) = \begin{cases} 1 & \text{if } f = 0 \\ 0 & \text{else.} \end{cases}$
- The loop state is by construction diff invar.

$$S : S^1 \rightarrow S^1 \quad S : W(f) \mapsto \alpha_S(W(f)) = W(Sf) \quad \alpha : \text{Diff}(S^1) \rightarrow \text{Aut}(\mathcal{W})$$

- The Fock rep has a phase anomaly, removable when $D=26$

$$U(S1)U(S2) = e^{i(S1 \circ S2)} U(S1 \circ S2)$$

- This means. $\omega_P(W(f))$ requires no crit dim, no tachyon
- However, is discontinuous and leads to unphysical results in non-gauge systems

Quantum Geometry from Topological String Theory

- Twisted $N=(2,2)$ SCFT $\Rightarrow$ Type A, B

A model on $X \leftrightarrow$ Kähler moduli of X ,

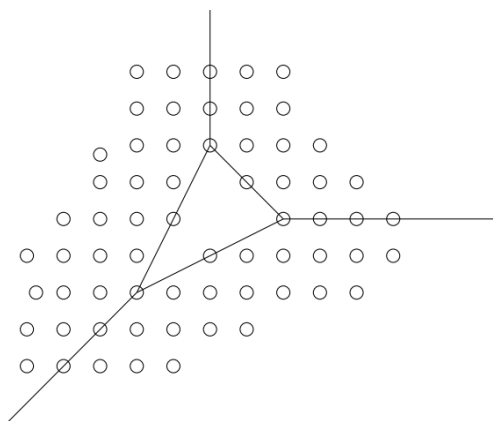
B model on $X \leftrightarrow$ complex moduli of X .

$$\mathcal{S} = \frac{1}{g_s^2} \int_X \frac{1}{3!} k \wedge k \wedge k .$$

- Counts holo. Sections (index)

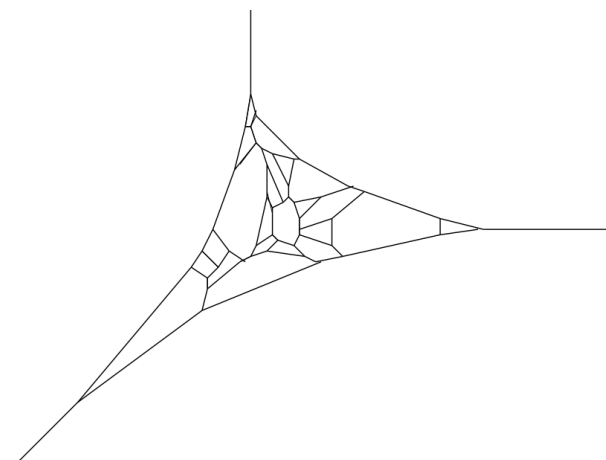
$$Z = \sum_k e^{\mathcal{S}} = \sum_k e^{N g_s}$$

- Quantum Geo from C^3 (Toric CY3) (Dijkgraaf '04)
- Blow up singular point with line bundle

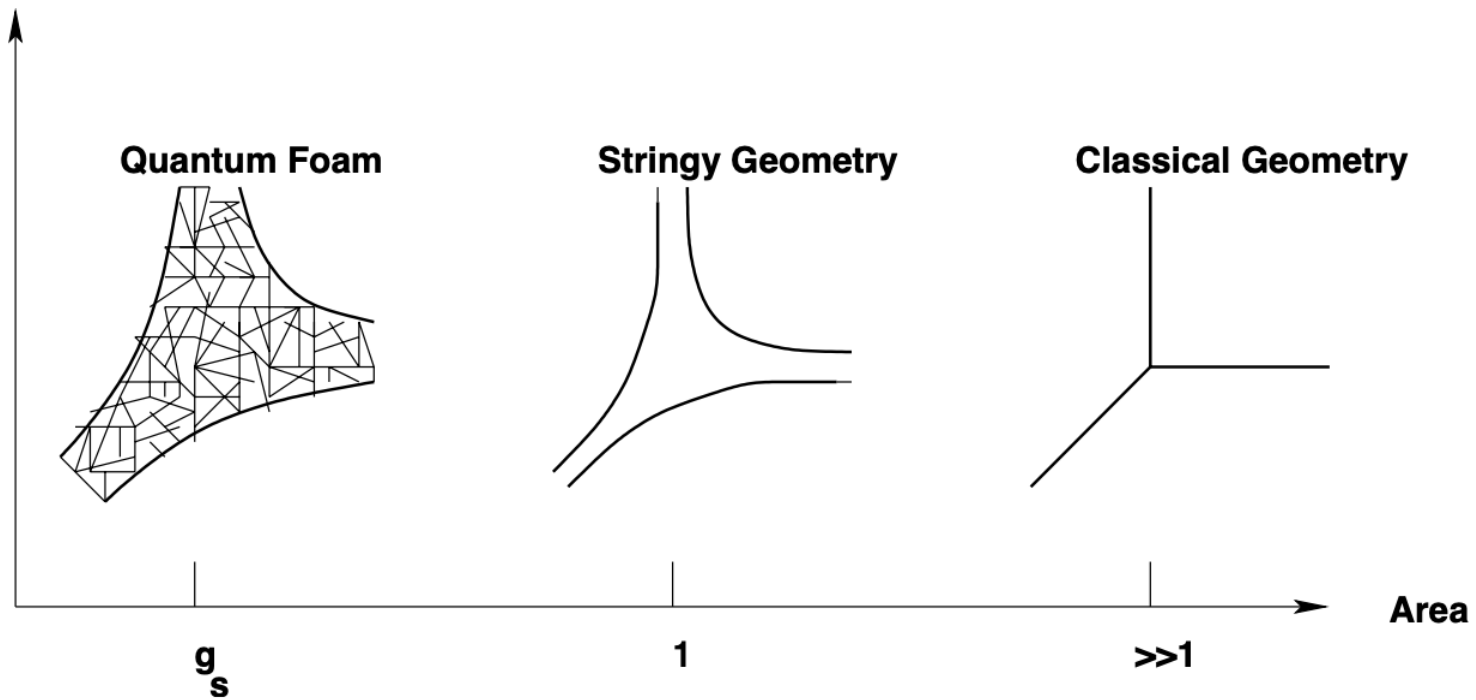


$$k = F g_s$$

$$Z = \sum_{\text{quantized toric blow ups}} \exp(N g_s)$$



Full geometry (Iqbal '03)



Exact partition function:

$$Z = e^F = \prod_{n \geq 0} \left(1 - e^{-a} q^{n + \frac{1}{2}} \right)$$

Difference equation:

$$Z(a + g_s, g_s) = \left(1 - e^{-a} q^{\frac{1}{2}} \right) Z(a, g_s)$$

Long Wavelength Topological M theory

- 7D G2 holonomy manifold defined by stable 3 form Φ

$$d\Phi = 0 \quad G = *\Phi \quad d*\Phi = 0$$

$$\Phi = \text{Re } \Omega + k \wedge dt, \quad G = \text{Im } \Omega \wedge dt + \frac{1}{2} k \wedge k$$

- Hitchin actions: M theory

$$V_7(\Phi) = \int_X \Phi \wedge *\Phi.$$

$$\text{Type A: } V_S(\sigma) = \frac{1}{6} \int_M k \wedge k \wedge k. \quad \text{Type B+B: } V_H(\rho) = -\frac{i}{4} \int_M \Omega \wedge \bar{\Omega}$$

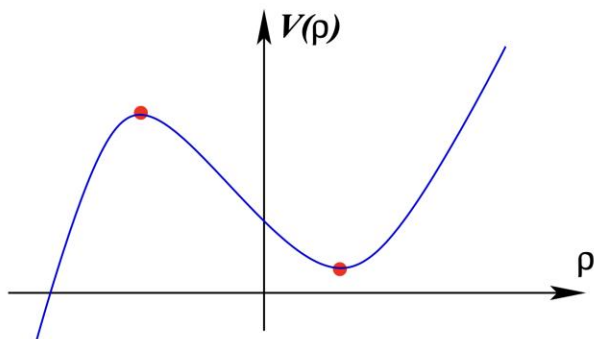


Fig. 1: Critical points of volume functionals define special geometric structures on X .

Metric	Holonomy	n	SUSY	Invariant p -forms
Calabi-Yau	$SU(3)$	6	1/4	$p = 2$: K (Kähler) $p = 3$: Ω
Exceptional	G_2	7	1/8	$p = 3$: Φ (associative) $p = 4$: $*\Phi$ (coassociative)
	$Spin(7)$	8	1/16	$p = 4$: Ψ (Cayley)

Top. M theory $\Rightarrow$ LQG top. sector

- 7D G2 man. $\Rightarrow R^4 \times M$, M is co-associative cycle
- SU(2) gauge field $\alpha^i = D_A y^i = dy^i + (A y)^i$

$$\Phi = e^{567} + e^5 \wedge (e^{12} - e^{34}) + e^6 \wedge (e^{13} - e^{42}) + e^7 \wedge (e^{14} - e^{23})$$

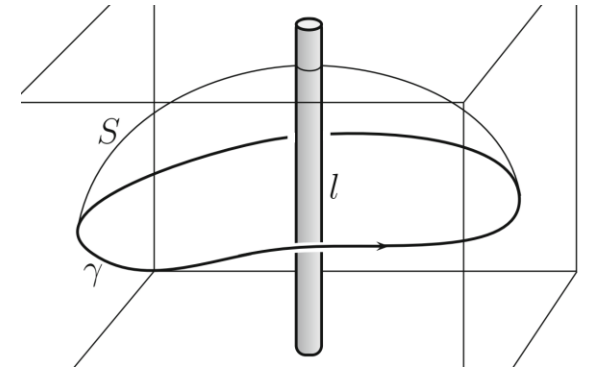
$$= \alpha^{123} + \alpha^1 \wedge \Sigma^1 + \alpha^2 \wedge \Sigma^2 + \alpha^3 \wedge \Sigma^3$$

$$*_\Phi \Phi = -\frac{1}{6} \Sigma_i \wedge \Sigma^i + \frac{1}{2} \epsilon^{ijk} e^i \wedge e^j \wedge \Sigma^k$$

- Metric: $\sqrt{g} g_{jk} = -\frac{1}{144} \Phi_{ji_1 i_2} \Phi_{ki_3 i_4} \Phi_{i_5 i_6 i_7} \epsilon^{i_1 \dots i_7} = -\frac{1}{12} \Sigma_{aa_1}^i \Sigma_{ba_2}^j \Sigma_{a_3 a_4}^k \epsilon^{ijk} \epsilon^{a_1 a_2 a_3 a_4}$
- E.O.M: $d\Phi = 0 \quad d *_\Phi \Phi = 0 \Rightarrow D_A \Sigma = 0 \quad F \wedge \Sigma = 0, \quad F_+ = \frac{\Lambda}{12} \Sigma$
- Full LQG: $S_{4D} = \int_{M^4} \Sigma^k \wedge F_k - \frac{\Lambda}{24} \Sigma^k \wedge \Sigma_k + \Psi_{ij} \Sigma^i \wedge \Sigma^j$
- LQG EOM: $F_i = \Psi_{ij} \Sigma^j + \frac{\Lambda}{12} \Sigma_i \quad D_A \Sigma = 0,$

H duality (Addazi, Marciano '18)

- (Bianchi '14) Gravitational Analog of Aharonov -Bohm
- Can one construct LQG from a classical system with finite d.o.f?
- Line defect carries $SU(2)$ mag flux recovered Holonomy-flux algebra
- Spacelike M-Branes charged under Φ are natural candidates for the line defect
- The Ham. constraint then can be seen from the string perspective as nucleation and decay of SM branes
- Correspondence not one to one



The logo for ASPER (Astrophysics and Space Physics Education Resource) is located on the left side of the slide. It features the word "ASPER" in a stylized, blue, serif font with a red outline. Above the letter "P" is a small blue starburst. The text is partially enclosed by a blue, curved, ribbon-like shape that loops around it.

More approaches

- Loop Quantum Super Gravity (Elder, Sahlmann '22)
- Loop Quantizing D-Branes, Open Strings, arbitrary target spaces in both the Bosonic and Fermionic (super) string
- Connection between $U(1)^3$ Euclidean gravity and twistor-string theory? (Bakhoda '22)(Abou-Zeid, Hull '06)
- Loops to study BMS ? (Campiglia '14)



Thank you!
Comments, Questions
and Discussions are
Welcomed!

Citations

- Eder, Konstantin, and Hanno Sahlmann. "Toward black hole entropy in chiral loop quantum supergravity." *Physical Review D* 106.2 (2022): 026001.
- Bakhoda, Sepideh. *The $U(1)^3$ Model of Euclidean Quantum Gravity*. Diss. Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), 2022.
- Abou-Zeid, Mohab, and Christopher M. Hull. "A chiral perturbation expansion for gravity." *Journal of High Energy Physics* 2006.02 (2006): 057.
- Campiglia, Miguel. "Note on the phase space of asymptotically flat gravity in Ashtekar–Barbero variables." *Classical and Quantum Gravity* 32.14 (2015): 145011.
- Bianchi, Eugenio. "Loop quantum gravity a la Aharonov–Bohm." *General Relativity and Gravitation* 46 (2014): 1-23.
- Addazi, Andrea, and Antonino Marciano. "Self-dual formulation of gravity in topological M-theory." *International Journal of Geometric Methods in Modern Physics* 17.03 (2020): 2050047.
- Dijkgraaf, Robbert, et al. "Topological M-theory as unification of form theories of gravity." *Advances in Theoretical and Mathematical Physics* 9.4 (2005): 603-665.
- Iqbal, Amer, et al. "Quantum foam and topological strings." *Journal of High Energy Physics* 2008.04 (2008): 011.
- Thiemann, Thomas. "The LQG string—loop quantum gravity quantization of string theory: I. Flat target space." *Classical and Quantum Gravity* 23.6 (2006): 1923.
- Helling, Robert C., and Giuseppe Policastro. "String quantization: Fock vs. LQG representations." *arXiv preprint hep-th/0409182* (2004).
- Helling, Robert C. "Lessons from the LQG String." *arXiv preprint hep-th/0610193* (2006).
- Donà, Pietro, and Simone Speziale. "Introductory lectures to loop quantum gravity." *arXiv preprint arXiv:1007.0402* (2010).